

Program : Diploma in Civil Engineering and Planning	
Course Code: 4399	Course Title: Advanced CAD Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To explain the basic concept of electrification, water supply and sanitary systems of a building and reproduce them as working drawings in CAD.
- To impart the knowledge of CAD commands for 3D building modelling and 3D rendering of various building components required for different civil engineering applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic construction details		Building materials and construction	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Develop rain water harvesting, septic tank drawing and service plan of the building & electrical layout of residential building.	15	Applying
CO2	Develop detailed 3D drawing of objects	8	Applying
CO3	Develop detailed 3D drawing of buildings	9	Applying
CO4	Applying rendering, shading animation effects	9	Applying
	Series Test	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	2		
CO2	3			3	2		
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Develop rain water harvesting, septic tank drawing and service plan of the building & electrical layout of residential building.		
M1.01	Develop detailed drawing of rain water harvesting	3	Applying
M1.02	Develop detailed drawings of septic tank	3	Applying
M1.03	Develop service plan of the building	3	Applying
M1.04	List out symbols used in electrical layout	3	Applying
M1.05	Draw detailed electrical layout of a residential building.	3	Applying
CO2	Develop detailed 3D drawing of objects		
M2.01	Apply various commands and develop simple shapes	4	Applying
M2.02	Develop simple 3D objects and obtain different views	4	Applying
	Series Test I	2	
CO3	Develop detailed 3D drawing of buildings		
M3.01	Develop building wire frame modeling	3	Understanding
M3.02	Develop 3D views of the building (Eg: Residential building, Library hall, Town hall, School building, hospital building etc.)	3	Applying

M3.03	Set a standard paper size with suitable title block using paper space and model space.	3	Applying
CO4	Applying rendering, shading animation effects		
M4.01	Develop simple 3D objects and apply rendering	3	Understanding
M4.02	Develop building models and apply rendering	3	Applying
M4.03	Apply modification of rendered images and animation effects to the model	3	Applying
	Series Test II	2	

Text / Reference:

T/R	Book Title/Author
T1	B. T. S. Prabhu, A. Achuthan - Building Drawing and Detailing
T2	S. P. Arora, S. P. Bindra - The text book of building construction
T3	W. B. McKay - Building construction

Online Resources

Sl. No	Website Link
1	https://archive.org/details/W.B.McKay

